MILITARY STANDARD

CARLOADING

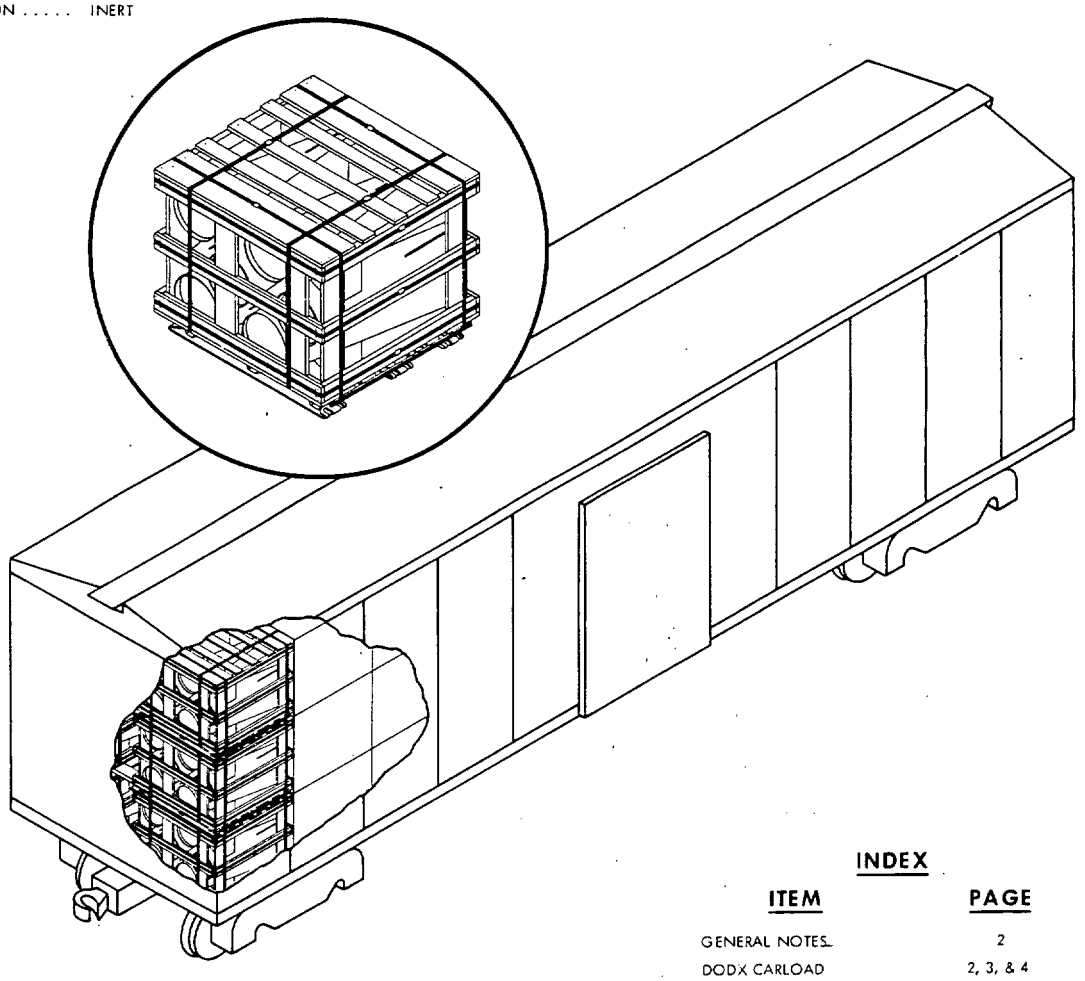
FIN ASSEMBLY, BOMB FOR 1000 LB MK 83 BOMB
(UNCRATED)
FLEET ISSUE UNIT LOAD

MIL-STD-1325-156
(NAVY)

19 SEPTEMBER 1972

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/222
WEIGHT . . . . . 926 LBS
DIMENSIONS . . . . 48 L x 45 1/2 W x 38 1/2 H
CUBE . . . . . 48.6 CU. FT.
DOT HAZARD CLASSIFICATION . . . . . INERT



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GENERAL NOTES	2
DODX CARLOAD	2, 3, & 4
40' 6" CARLOAD, COM'L	6 & 7
50' 6" CARLOAD, COM'L	8, 9, & 10

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO COMPREHENSIVE INDEX NWHL NO. 4200.

AUTHORIZED AND RELEASED FOR GENERAL USE.

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE (AIR) SYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

NOT APPLICABLE

DATE

ORIGINATOR

SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 70000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 IN. INSIDE LENGTH, 107 3.4 INCHES INSIDE WIDTH BETWEEN RAILS (111 IN. INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 IN. INSIDE LENGTH, 110 IN. INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 IN. INSIDE LENGTH, 110 IN. INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF BOMB FINS FOR 1600 LB L.D. BOMBS UNCRATED PALLETIZED IN ACCORDANCE WITH WR-54/222.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC MM-L-751
   NAILS - FED. SPEC FF-N-105
10. AFTER BLOCKING AND BRACING HAVE BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA AND CLOSE AND SEAL BOXCAR DOORS.

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 68 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH. SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.

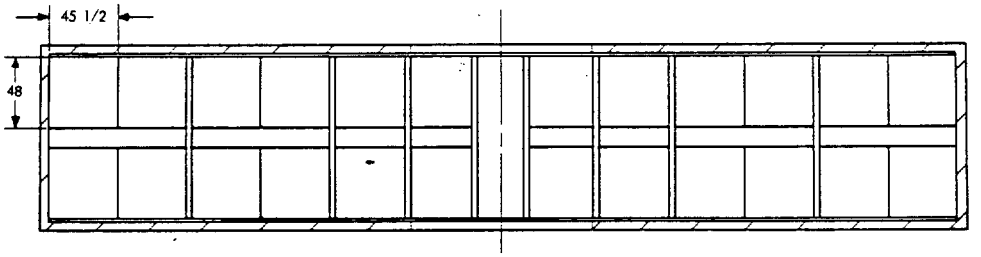
*W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR.

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

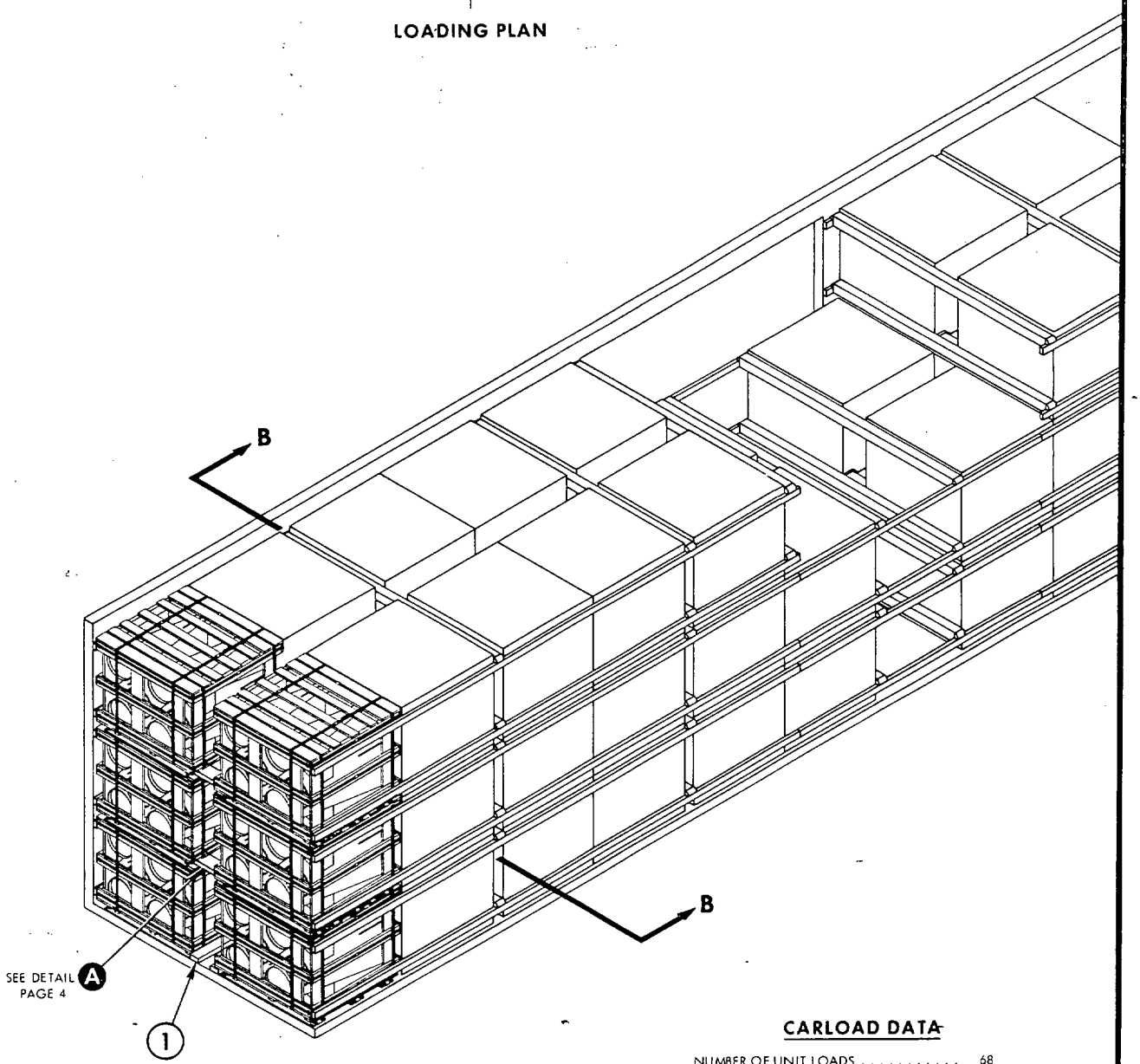
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
4	CROSS BRACE	2 x 6 x W*	66	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 45	44	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	44	SEE 3	-	-
1	SLEEPER	2 x 4 x 45	24	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

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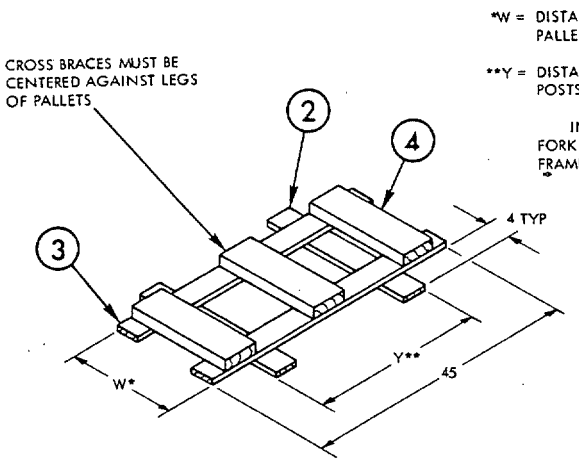
LOADING PLAN



CARLOAD DATA

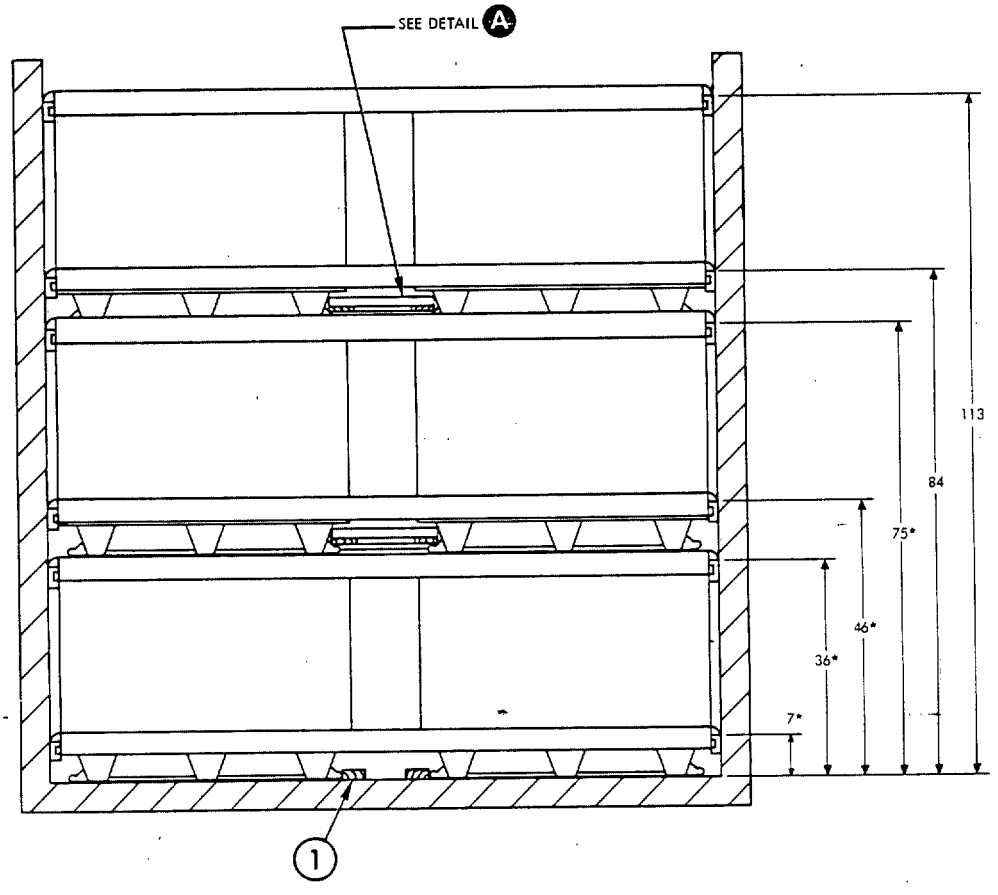
NUMBER OF UNIT LOADS	68
NUMBER OF CROSS MEMBERS REQ'D	36
LOAD WEIGHT	62,968 LBS
DUNNAGE WEIGHT	624 LBS
CARLOAD WEIGHT	63,592 LBS

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DETAIL A

SWAY BRACE ASSEMBLY
24 REQUIRED



SECTION B-B

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSSMEMBERS

* DOORWAY MEMBER LOCATIONS

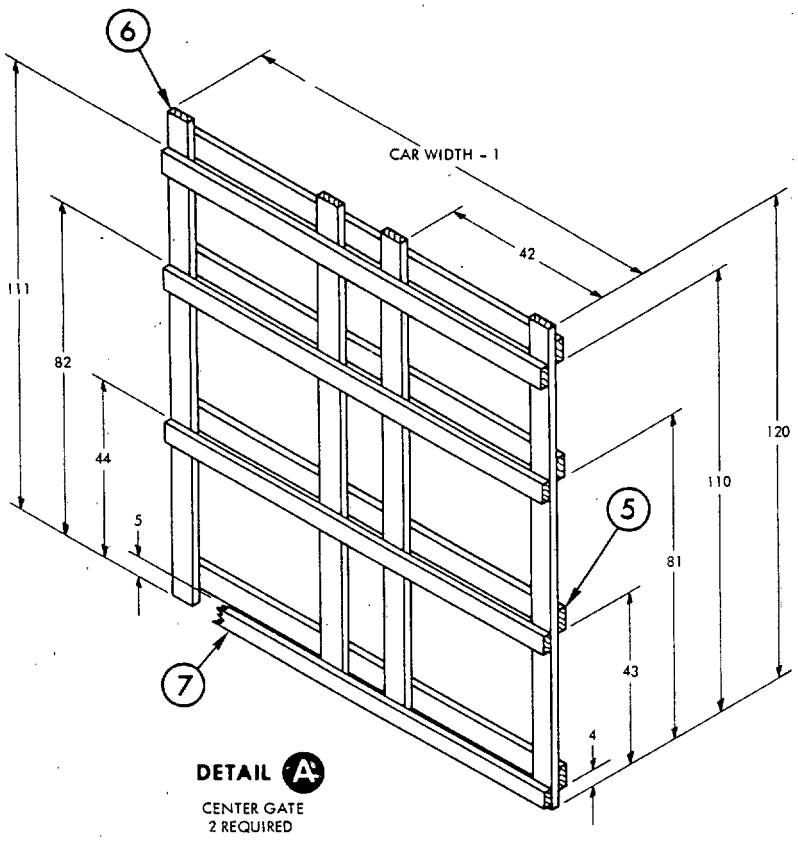
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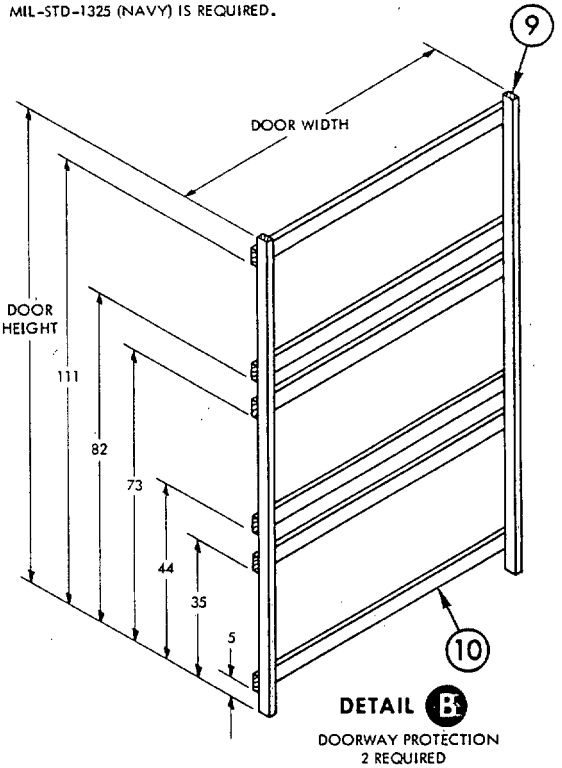
40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 60 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.



DETAIL A
CENTER GATE
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED.



DETAIL B
DOORWAY PROTECTION
2 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
10	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	11	2 PER JOINT	10d
9	VERTICAL DOORWAY MEMBER	2 x 3 x DOOR HEIGHT	4	DOOR POST	2 PER FOOT	20d
8	STRUT	4 x 4 x WEDGE FIT ***	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 120	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W *	60	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 45	40	2, 4	SEE NOTE ***	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8) *	40	SEE 3	-	-
1	SLEEPER	2 x 4 x 45	20	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

* W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR.

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

*** TOP LAYER STRUTS NEED NOT BE WEDGE FIT AND SHOULD BE TOE NAILED TO PIECE 6 WITH TWO 10d NAILS ON EACH ACCESSIBLE SIDE. 2 x 6 STRUTS, DOUBLED AND LAMINATED WITH 10d NAILS, MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

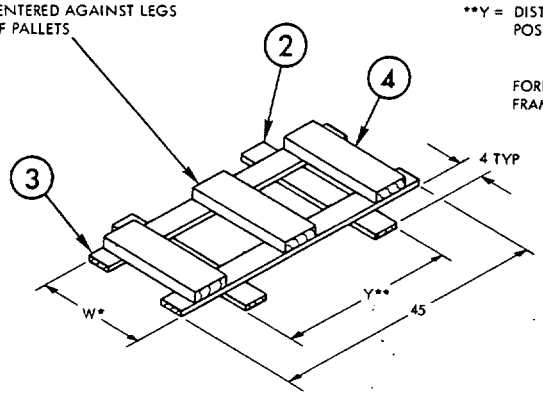
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*W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

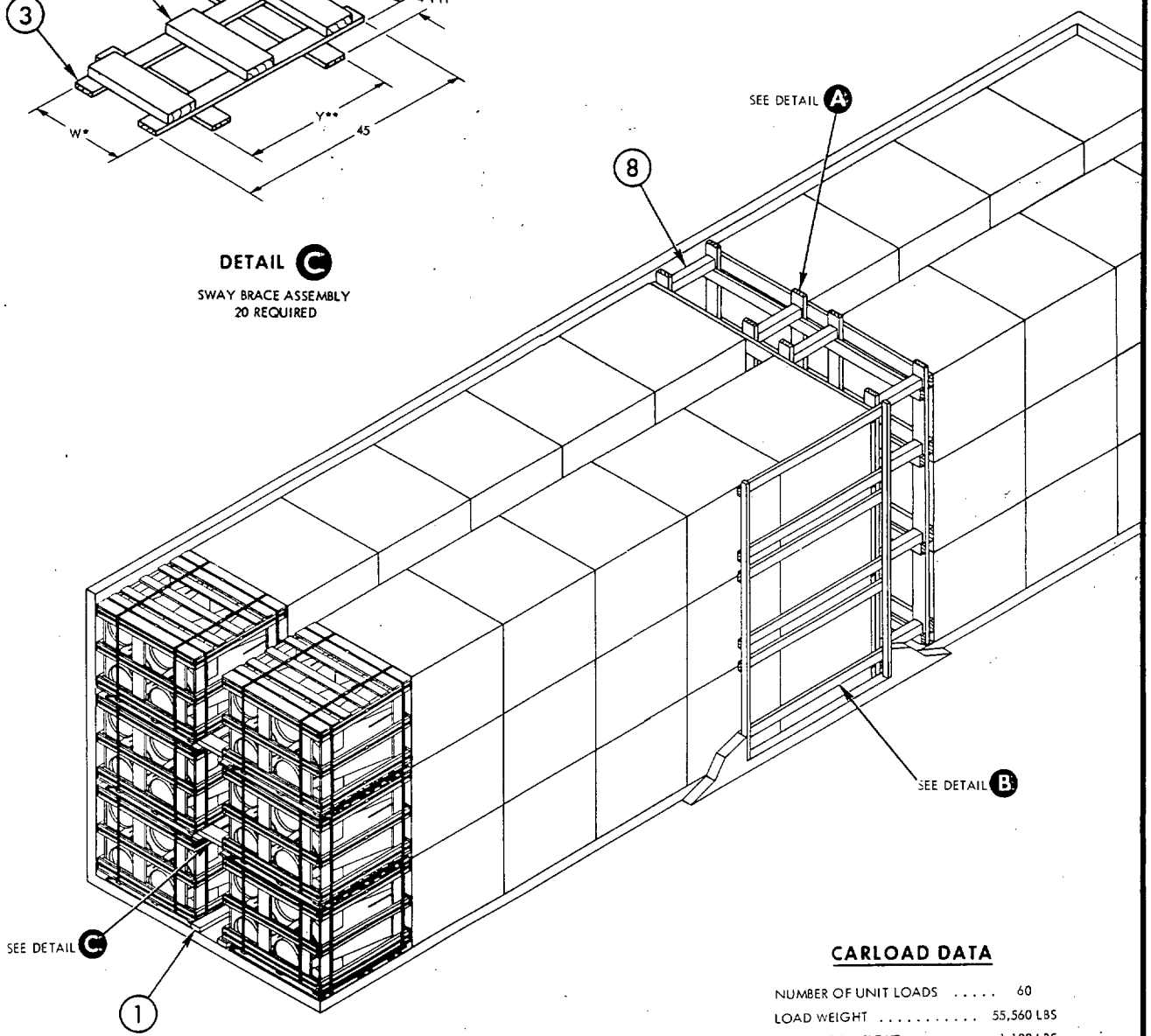
INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS



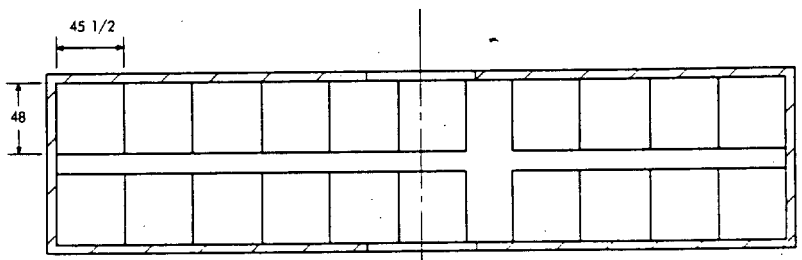
DETAIL C

SWAY BRACE ASSEMBLY
20 REQUIRED



CARLOAD DATA

NUMBER OF UNIT LOADS	60
LOAD WEIGHT	55,560 LBS
DUNNAGE WEIGHT	1,102 LBS
CARLOAD WEIGHT	56,662 LBS

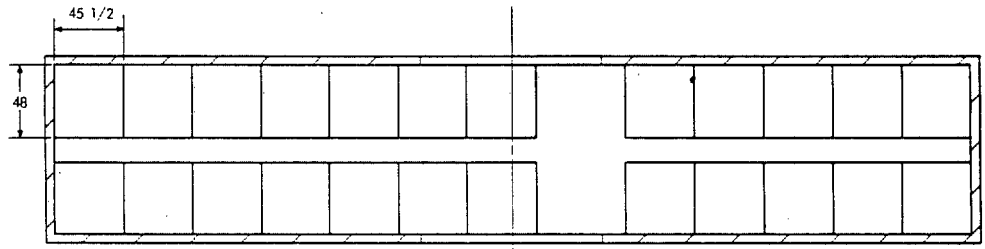


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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 72 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL D) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 9 AND 10 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
10	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	11	2 PER FOOT	10d
9	VERTICAL DOORWAY MEMBER	2 x 3 x DOOR HEIGHT	4	DOOR POST	2 PER FOOT	20d
8	STRUT	4 x 4 x WEDGE FIT ***	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 120	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W *	72	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 45	48	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8) *	48	SEE 3	-	-
1	SLEEPER	2 x 4 x 45	24	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

* W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR.

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

*** TOP LAYER STRUTS NEED NOT BE WEDGE FIT AND SHOULD BE TOE NAILED TO PIECE 6 WITH TWO 10d NAILS ON EACH ACCESSIBLE SIDE. 2 x 6 STRUTS, DOUBLED AND LAMINATED WITH 10d NAILS, MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

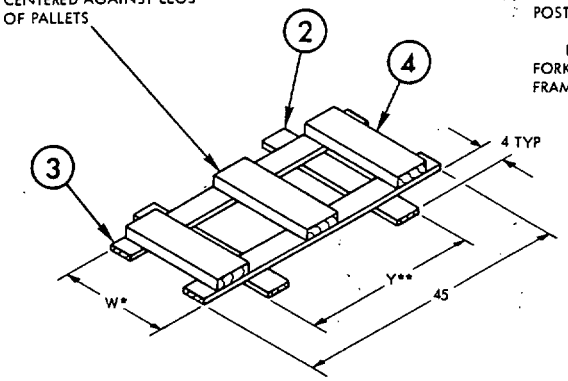
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**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

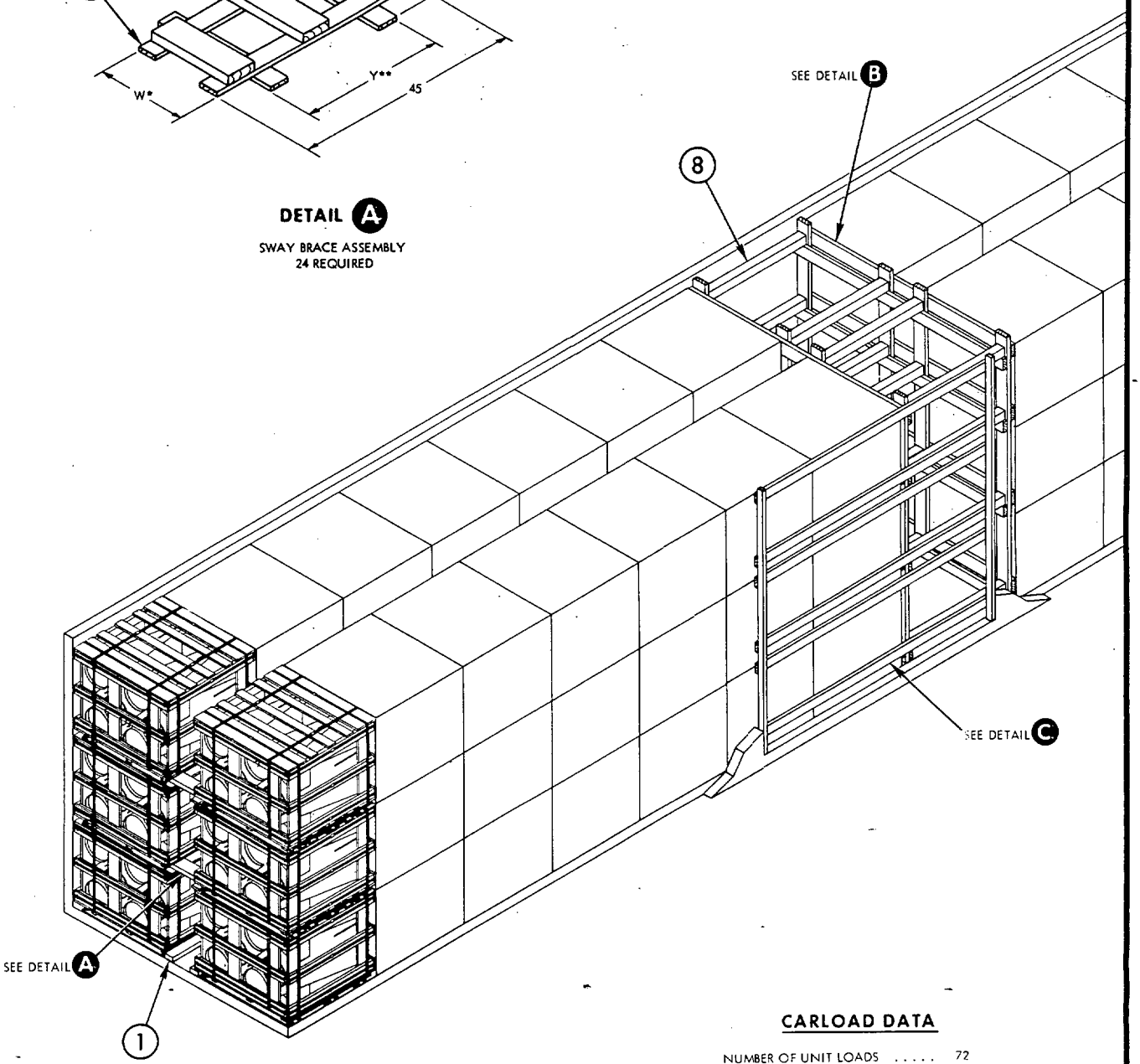
INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS



DETAIL A

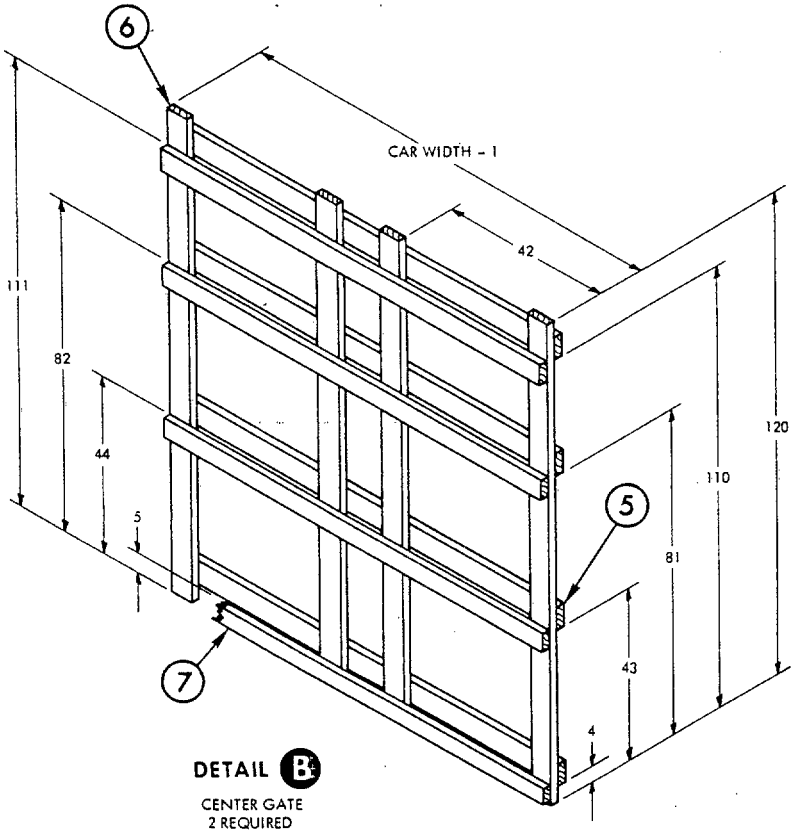
SWAY BRACE ASSEMBLY
24 REQUIRED



CARLOAD DATA

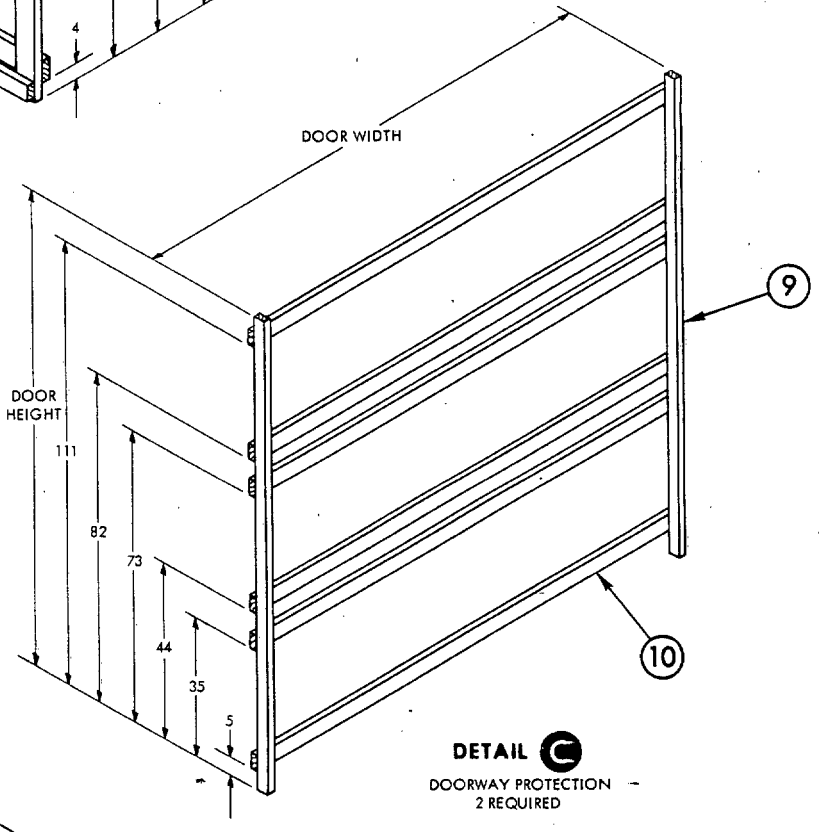
NUMBER OF UNIT LOADS	72
LOAD WEIGHT	66,672 LBS
DUNNAGE WEIGHT	1,625 LBS
CARLOAD WEIGHT	68,297 LBS

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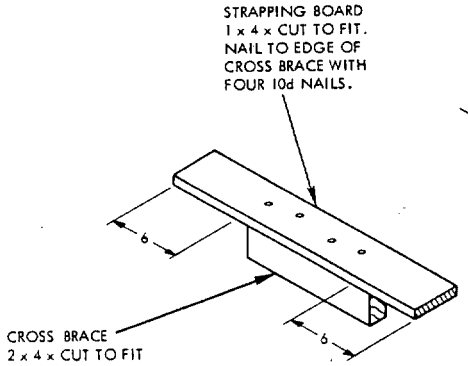


DETAIL B
CENTER GATE
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED.



DETAIL C
DOORWAY PROTECTION
2 REQUIRED



DETAIL D
TOP SWAY BRACE

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